YUANJUN (JOHNNY) DAI

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EDUCATION

Case Western Reserve University 08/2019 – Dec. 2026 (exp.)
Ph.D. Candidate in Computer Science · Advisor: Prof. An Wang **(NSF CAREER Award)** Cleveland, OH
Research focus: in-kernel system observability, GPU telemetry for AI/ML workloads, and distributed system monitoring

University of Pittsburgh 2014
M.S. in Electrical and Computer Engineering Pittsburgh, PA
Research focus: Computer Architecture and Microarchitecture

Northeastern University 2008
B.S. in Electrical Engineering Shenyang, China

PUBLICATIONS

[1] ***Yuanjun Dai**, K. He, A. Wang. "DYNAMIX: RL-Based Adaptive Batch Size Optimization in Distributed Machine Learning Systems." *IEEE IPDPS Workshop (PDCO)*, 2026.

[2] ***Yuanjun Dai**, Q. Guo, X. Wang. "PSketch: A Priority-Aware Sketch Architecture for Real-Time Flow Monitoring via eBPF." *IEEE ICNC*, 2026.

[3] ***Yuanjun Dai**, Q. Guo, X. Wang. "eHashPipe: Lightweight Top-K and Per-PID Resource Monitoring with eBPF." *IEEE CCNC*, 2026.

[4] ***Yuanjun Dai**, A. Wang, Y. Guo, S. Chen. "Elastically Augmenting the Control-Path Throughput in SDN to Deal with Internet DDoS Attacks." *ACM Transactions on Internet Technology (TOIT)*, Vol. 23(1), 2023.

[5] ***Yuanjun Dai**, Q. Guo, A. Wang. "DNN Architecture Attacks via Network and Power Side Channels." *Springer SecureComm*, 2023.

[6] ***Yuanjun Dai**, Q. Guo, A. Wang. "Stealing DNN Architectures via Network and Power Side Channels." *ACM Transactions on Privacy and Security (TOPS)* — Under Review.

[7] Q. Guo, **Yuanjun Dai**, A. Wang et al. "PRISM: Priority-Based Selective Reliability for Efficient Distributed Machine Learning." Under Review.

RESEARCH EXPERIENCE

Research Assistant 08/2019 – Dec. 2026 (exp.)
Case Western Reserve University

In-Kernel eBPF Sketch Systems for DML Network and System Resource Monitoring | Paper (PSketch) / Code · Paper (eHashPipe) / Code

- **PSketch**: Designed a three-tier in-kernel network monitoring system in **eBPF** that classifies flows by behavior — priority flows get exact per-packet tracking via a hashmap, elephant flows are identified and tracked via a heavy flow table with **voting-based kick-out collision resolution**, and remaining mice flows are approximated via a **3-layer Count-Min Sketch**; achieves **O(n) Top-K detection**, **96% accuracy**, and <1.1% throughput overhead at 10 Gbps
- **eHashPipe**: Designed a dual-pipeline in-kernel observability framework in **eBPF** for DML workloads — one pipeline profiles memory usage via HashPipe with cascading slot eviction and deallocation tracking, the other tracks per-thread CPU time by hooking the **sched_switch** tracepoint; delivers **10–14× finer temporal resolution** than top/htop at ~11ms latency

Scalable AI Model Serving Infrastructure on Kubernetes | Code / Video

- Designed a self-hosted LLM inference platform on **Kubernetes** (NGINX → **FastAPI** → **vLLM**) with **GitOps CI/CD** (**ArgoCD**), **GPU time-slicing**, hardware **MIG** partitioning, and custom-metrics **HPA** autoscaling
- Built 4-layer observability via **Prometheus** + **Grafana**: GPU hardware (**DCGM**: utilization, VRAM, temperature, power), **vLLM** inference (queue depth, KV cache, TTFT), gateway/router request counters and latency histograms per endpoint (**prometheus_client**), and K8s cluster metrics (**kube-prometheus-stack**)

DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML | Paper / Code

- Designed a **PPO**-based **Reinforcement Learning (RL)** scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size, evaluated on **PyTorch DDP** and **Parameter Server (BytePS)**
- Achieves **46% wall-clock speedup** and **~6% accuracy gain** across **8–64 A100 GPUs** over fixed batch size baseline

PRISM – Priority-Based Selective Reliability for Distributed ML Training | Paper

- Designed a transport-layer library (**NCCL/Gloo → UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

Scotch – Elastic Distributed Overlay for Control-Plane Scalability and Load Balancing | Paper / Code

- Designed an **OvS**-based elastic overlay: when physical switch OFA is saturated, tunnels new flows to a **vSwitch** pool via **event-driven** Packet-In Edge messages; built a centralized **Ryu SDN** controller for **flow monitoring** and **load balancing** across vSwitches
- Separates small disruptive flows (terminated at vSwitches) from large flows (migrated to physical paths); achieves **>70% controller congestion reduction** on 34-node GENI Clos topology without hardware modification

AI-Powered Job Hunting Agent | Code / Report

- Built a **multi-agent AI platform** using **LangGraph** for automated job discovery and resume optimization — orchestrates specialized agents for collection, scoring, and refinement across 5 platforms, with a **6-dimension LLM matching framework** designed via **prompt engineering**
- Implemented **capability-aware multi-model routing** (**DeepSeek** for high-volume scoring, **Claude Opus** for quality-critical tailoring) and a **FastAPI** backend with cross-platform deduplication and health monitoring endpoints

LLM Fine-tuning for Multi-label Text Classification on Imbalanced Data | Paper

- Built a multi-label text classifier across 115 highly imbalanced labels — applied **inverse-frequency weighted loss** and **per-label threshold calibration** to improve long-tail performance
- Fine-tuned **PubMedBERT** via **PyTorch DDP** on 2×RTX A6000 GPUs; fine-tuned **Qwen-7B** via **NeMo + Megatron-LM** with **FSDP (ZeRO-3)** and **FlashAttention** to fit under GPU memory constraints; used **few-shot prompting** to mitigate hallucination in label generation

Side-Channel Attack Analysis and Behavioral Profiling of Distributed ML Systems | Paper / Code

- **Analyzed DML computation/communication behavior** in **MXNet BSP** Parameter Server: instrumented push/pull primitives and modified MXNet source code to capture **DML network behavior** and ground-truth per-layer traffic patterns
- Profiled **ML energy consumption** via **Intel RAPL**; used **band-pass filtering** and **K-means clustering** for noise reduction and layer segmentation; applied **polynomial regression** for **activation function energy profiling**; classified layer types via **Decision Tree** and **MLP ensemble**

WORK EXPERIENCE

Software Engineer II
06/2015 – 08/2016

Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK; contributed to packet forwarding modules, reducing latency under heavy load.

Software Engineer Intern
10/2014 – 05/2015

Broadcom (Emulex)

- Re-architected RTOS data acquisition (DMA + circular-buffer, **30% CPU reduction**); ported BladeEngine ASIC CLI from Linux to Windows Server.

Technical Product Manager
07/2018 – 01/2019

Xiaomi

- Defined IoT appliance requirements via competitive analysis and user research; translated market needs into actionable engineering specifications.
- Assessed technical feasibility and delivered architecture proposals covering trade-offs and risks to align engineering and business stakeholders.

Product Manager
09/2016 – 06/2018

Midea Group

- Led technical PM for **Fun Oven II** across ML model training, Docker deployment on Alibaba Cloud, mobile app, and culinary algorithm design; won **Red Dot Design Award**, debuted at 2018 AWE Show · Red Dot ↗ · PR Newswire ↗

SKILLS

Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Languages	Python, C/C++
ML Frameworks	PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Tensor Parallel (TP), Pipeline Parallel (PP), Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines (GitHub Actions), ArgoCD, Slurm
Systems & APIs	FastAPI, REST API, gRPC
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)
Agentic AI	LangGraph, Multi-Agent Orchestration, Multi-Model Routing, Prompt Engineering

PATENTS

Oven Temperature Control Method, Device and Computer-Readable Storage Medium — Granted CN 108594898 B | Dai Yuanjun et al.